

Problem 4

[Your Name / Group Name]

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1 Benchmarks

To verify the performance and robustness of the Proposed Heuristic, this experiment selected the following two benchmarks for comparison:

1. **Optimal Solution:** By limiting the instance size to a computationally tractable range, we use a mathematical programming solver to solve the problem. This provides the true optimal objective value as an absolute performance baseline.
2. **Pure Greedy Algorithm:** Served as a simple and intuitive baseline. The steps of this algorithm are as follows:
 - **Step 1:** Sort all orders in descending order based on "Absolute Revenue". If the revenue is identical, prioritize the order with an earlier pickup time.
 - **Step 2:** Iterate through the sorted orders. For each order, search outward from the pickup station (in increasing order of relocation time).
 - **Step 3:** Find the nearest available vehicle that satisfies the level and time window constraints. If found, and the remaining relocation budget allows, assign it immediately and deduct the corresponding budget and vehicle availability time; if no vehicle is available, reject the order and evaluate the next one.

2 Experimental Design

To comprehensively evaluate the algorithm's performance under various business scenarios, we designed stress tests covering spatial, temporal, and resource dimensions, categorizing them into 7 distinct scenarios.

2.1 Fixed Parameters

Across all scenarios, the base environment settings are as follows:

- Number of Stations ($|S|$): 20
- Number of Cars ($|C|$): 100

- Number of Car Levels ($|L|$): 3 (Hourly Rates: Level 1 = 400, Level 2 = 800, Level 3 = 1200)
- Total Number of Orders ($|K|$): 200
- Planning Horizon (D): 7
- Relocation Budget (B): 6000

2.2 Test Scenarios and Factors

By adjusting factors such as order distribution, relocation budget, and fleet structure, we established the following 7 test scenarios. 100 random instances were generated for each scenario. Note that any parameter not explicitly mentioned in Scenarios 02–07 follows the baseline settings of Scenario 01 unless otherwise specified.

Table 1: Description of Experimental Scenarios

Scenario	Description & Parameter Distribution
01	Order time, duration, stations, and levels are uniformly distributed.
02	80% of orders request Level 1, but the initial fleet consists mostly of high-level cars (L1: 10%, L2: 40%, L3: 50%).
03	From day 1 to day 3, 70% of the total orders are bad orders (fixed at Level 1) requiring long vehicle occupation (24–48 hrs). From day 4 to day 7, 30% of the total orders are good orders (fixed at Level 3) requiring short vehicle occupation (4–12 hrs).
04	80% of orders originate at stations 1 or 2, but return at the last two stations.
05	Order occurrences are uniform, but orders in the last 2 days have extremely long durations (24–48 hrs); the first 5 days have short durations (2–6 hrs).
06	The first 5 days contain only 30% of total orders, while the remaining 70% erupt in the last 2 days; duration remains random.
07	The relocation budget is severely restricted to $B = 2000$ (compared to the baseline $B = 6000$).

3 Results and Analysis

This section presents the Optimality Gap of the Proposed Heuristic and the Pure Greedy Algorithm relative to the Optimal Solution. The Gap is calculated as:

$$Gap = \frac{\text{Optimal} - \text{Heuristic}}{\text{Optimal}} \times 100\% \quad (1)$$

3.1 Average Performance and Standard Deviation

Table 2 summarizes the average gaps and standard deviations under each scenario.

Table 2: Statistics of Optimality Gaps and Standard Deviations by Scenario

Group Name (Scenario)	Avg Gap_H	Std Dev H	Avg Gap_G	Std Dev G
01_All_Random	0.01 %	0.09 %	6.98 %	5.87 %
02_Upgrade_Test	2.56 %	3.85 %	36.04 %	11.57 %
03_Value_Trap	4.19 %	3.14 %	53.46 %	7.62 %
04_Spatial_Imbalance	6.93 %	6.67 %	11.68 %	6.79 %
05_Late_Duration_Peak	0.00 %	0.00 %	9.87 %	6.10 %
06_Late_Quantity_Peak	0.68 %	1.27 %	4.70 %	3.67 %
07_Budget_Trap	5.80 %	8.64 %	15.69 %	7.41 %

3.2 Performance Visualization

Figure 1 illustrates the performance comparison across all scenarios.

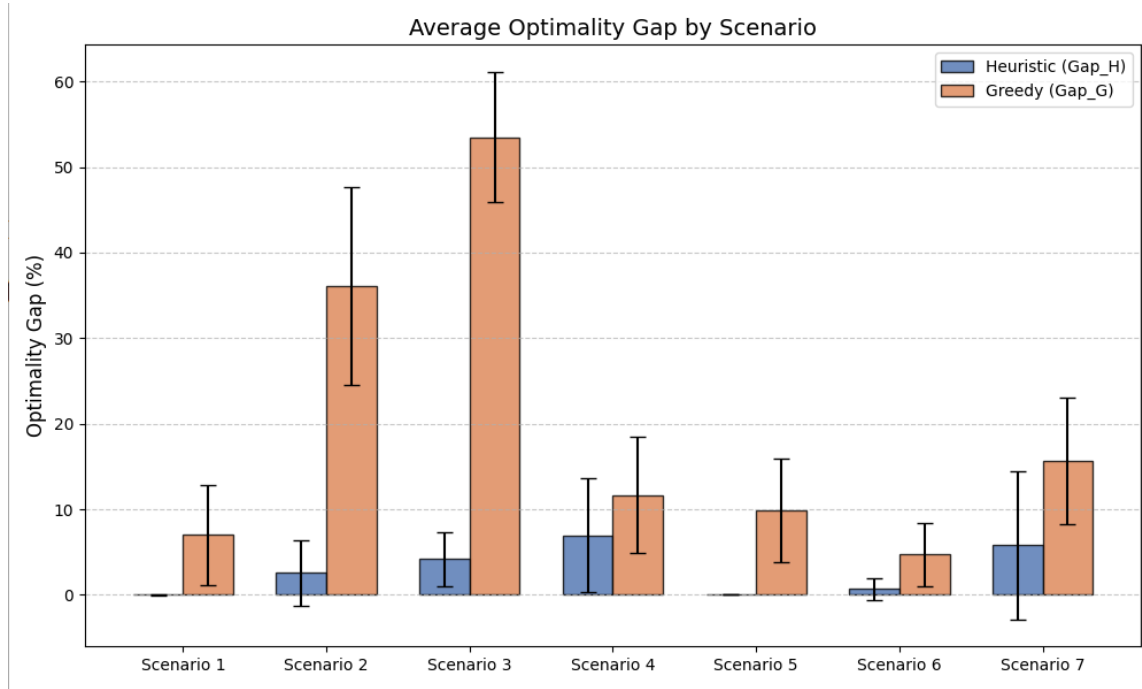


Figure 1: Average Optimality Gap by Scenario